

TraqOOH — Infrastructure Stack

Full-stack service map · costs · free limits · scale thresholds

What	Service	Cost	Free limit	When you'll hit it
HOSTING & FRONTEND				
Marketing site	Vercel	Free	100 GB/mo bandwidth 100 deploys/day	Never — static HTML, only hits with millions of visitors/mo
Web app frontend	Cloudflare Pages	Free	Unlimited bandwidth 500 builds/mo	Never on bandwidth. Builds only if deploying 500+ x/mo
BACKEND & DATABASE				
FastAPI backend	Render	Free (sleeps)	750 compute hrs/mo	Already near limit — upgrade to Starter (\$7/mo) to eliminate 50s cold starts
Database	Neon PostgreSQL	Free	0.5 GB storage 10 connections max	~1-2 years at current scale. Upgrade around 500+ campaigns or 0.4 GB storage
STORAGE & MEDIA				
Photo & video	Cloudflare R2	Free	10 GB storage 10M uploads/mo	~500 photos/mo = 20 months. Each photo ~0.5 MB after compression
DNS & NETWORKING				
DNS	Cloudflare	Free	Unlimited	Never
MOBILE APP				
Android app build	Expo EAS	Free	30 builds/month	Only if updating the app 30+ x/mo — normally 2-5 builds/mo
MONITORING & UPTIME				
Keep-alive (1 min)	cron-job.org	Free	Unlimited jobs Pings /health every 1 min	Never — Render stays warm permanently
Uptime monitoring	UptimeRobot	Free	50 monitors · 5 min check	Never — using 1 of 50 monitors
Error monitoring	Sentry	Free*	5,000 errors/mo 10K transactions/mo	Code deployed — set VITE_SENTRY_DSN in Cloudflare Pages env vars to activate
COMMUNICATIONS				
Email OTP & links	Resend	Free	3,000 emails/mo 100/day cap	~100 mobile logins/day hits daily cap. Fine for years at small-team scale

* Sentry: code deployed — set VITE_SENTRY_DSN in Cloudflare Pages env vars to activate.

Upgrade Triggers — When to Pay

Trigger	Action	Monthly cost
Users complaining about 50s load times (right now)	Upgrade Render Free → Starter	\$7/mo
DB connection errors under heavy use (Neon 10-connection limit exhausted)	Add Neon connection pooler (PgBouncer) in Neon settings — free	Free
DB storage approaching 0.4 GB	Upgrade Neon Free → Launch plan (10 GB storage, 7-day PITR, more connections)	\$19/mo
R2 storage exceeds 10 GB	Pay-as-you-go auto-activates — no action needed	\$0.015/GB/mo (very cheap)
Resend hitting 100 emails/day cap (100+ daily OTP logins from mobile users)	Upgrade Resend Free → Pro (50,000 emails/mo, no daily cap)	\$20/mo
Updating Android app more than 30x/month	Upgrade EAS Free → Production (unlimited builds, priority queue)	\$99/mo
Sentry exceeding 5,000 errors/month	Upgrade Sentry Free → Team (50,000 errors/mo, 90-day retention)	\$26/mo
Render cold starts return after heavy traffic or need zero-downtime deploys	Upgrade Render Starter → Standard	\$25/mo

Total Monthly Cost by Stage

Stage	Total cost	What changes
Right now (free tier)	\$0/mo	Cold starts only issue
Recommended before launch	\$7/mo	Render Starter — always-on
Early growth (10-50 clients)	\$26/mo	+ Neon Launch
Growth (50-200 clients)	\$46/mo	+ Resend Pro
Scale (200+ clients)	\$98/mo	+ Sentry Team

Only urgent action: Render Free → Starter (\$7/mo) — eliminates 50s cold starts before first real client.